

Random Graphs, Giant Components, and Preferential Attachment

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February 18, 2025

Abstract

This paper studies two crucial topics in random graph theory: the emergence of giant components in the configuration model and the preferential attachment mechanism underlying scale-free networks. We first review branching process heuristics and the sprinkling argument used to characterize the existence and uniqueness of giant components in sparse random graphs, including the Erdős–Rényi model and the configuration model. We then take a closer look at the Barabási–Albert framework, analyzing how preferential attachment produces heavy-tailed degree distributions and asymptotic power-law behavior. Together, these topics illustrate how probabilistic methods explain large-scale structural phenomena in network models.

1 Review of a Giant in the Configuration Model

In this section, let's review the existence of a giant component in the Configuration Model.

1.1 Assumptions

We will start by making 3 key assumptions.

1.)

$$\frac{1}{n} \sum_{i=1}^n \mathbf{1}_{d_i=k} \rightarrow P(D = k)$$

- If a node is chosen at random from the graph, $P(D = k)$ is the limiting probability that the node has degree k .

- $\mathbf{1}_{d_i=k}$ is an indicator variable that takes on the value 1 if node i has the degree k and 0 otherwise.

As the number of nodes n grows large, the proportion of nodes with degree k converges to $P(D = k)$.

2.)

$$\frac{1}{n} \sum_{i=1}^n d_i \rightarrow E[D]$$

- $E[D]$ is the expected degree of a randomly selected node in the graph.
- The sum of the degrees of all nodes divided by the number of nodes (i.e. the mathematical mean) converges to $E[D]$ as n grows large.

3.)

$$\frac{1}{n} \sum_{i=1}^n d_i^2 \rightarrow E[D^2]$$

1.2 Two-Stage Branching Process

The Two-Stage Branching Process helps us understand whether a large random graph contains a giant component. The process starts by choosing a node at random (we will refer to it as node1) in a large graph. Then, instead of picking another node at random, we will traverse a random edge connected to node1, arriving at node2. Therefore, as this process continues, it is evident that not all nodes are equally likely to be reached, and higher-degree nodes are, in fact, more likely to be reached. Thus, node i with degree d_i will be chosen with probability proportional to d_i .

Let's define d_i^* as the "excess degree" of a node. Excess degree represents the number of edges connected to the node excluding the edge that we traversed to get to arrive at the node. i.e.

$$d_i^* = d_i - 1$$

After we have reached a node, we typically want to know the number of available edges to continue exploration (excluding the edge we traversed to get to the node). Conditioning on these two effects, we see that the limiting degree distribution of d_i^* is given by:

$$P(d_i^* = k) = \frac{(k+1)P(d_i = k+1)}{\mathbb{E}[d_i]}$$

The expected excess degree ν is:

$$\nu = \mathbb{E}[d_i^*] = \frac{\mathbb{E}[d_i(d_i - 1)]}{\mathbb{E}[d_i]}$$

ν helps us determine if a giant component exists in a large random graph, since:

- if $\nu > 1$, a giant component exists, because a random node x will have more than 1 node connected to it (excluding the node whose edge we traversed to reach x).
- if $\nu \leq 1$, there is no giant component.

1.3 Theorem for Emergence of a Giant Component

In addition to the 3 assumptions made in section 1.1, let's assume $d_i \geq 1 \forall i$ and $P(D = 2) < 1$, then:

Theorem 0.1.

$$\frac{C_{\max}}{n} \xrightarrow{\mathbb{P}} \theta = P(T_{D,D^*} \text{ survives})$$

where $C_{\max}$ represents the number of nodes in the largest component of the random graph.

Technique to prove Theorem 9.1.

Step 1)

Define V_+ as:

$$V_+ = \{x \in [n] \mid |C(x)| > k_n\}$$

where:

- $|C(x)|$ is the total number of nodes in the component containing node x .
- k_n is the threshold to choose nodes with large components, and $k_n \rightarrow \infty$ ensures that the threshold increases as n grows large.

We want to show:

$$\frac{|V_+|}{n} \xrightarrow{\mathbb{P}} \theta > 0$$

which means that as the size of the graph n grows large, the proportion of nodes in large components converges (in probability) to some $\theta > 0$. Note that it seems reasonable to define $|V_+|$ as the set of nodes with high degree. However, the issue that arises is that a node with high degree does not necessarily belong to a giant component, as seen in Figure 1.

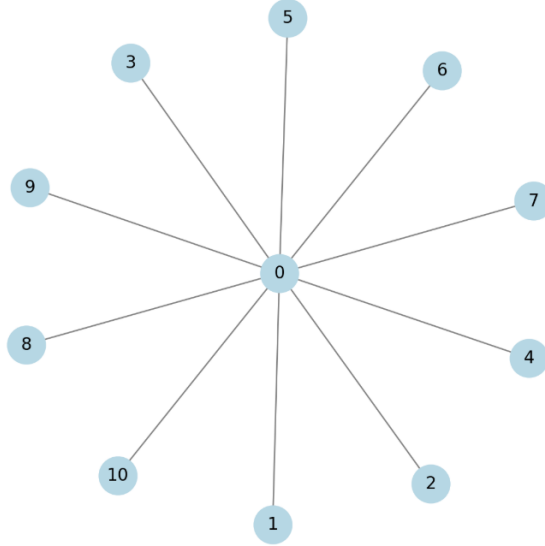


Figure 1: Node 0 has high degree but belongs to a relatively small component.

Step 2)

We now have V_+ , the set of nodes that belong to giant components. To complete the proof of Theorem 9.1, we can use Sprinkling to prove that $\forall \epsilon > 0$, all but ϵn nodes belong to a single component. We won't show how this works in $CM_n(d)$, but explain it for the simple case of $G(n, \frac{c}{n})$.

1.4 Sprinkling in $G(n, \frac{c}{n})$

In the $G(n, \frac{c}{n})$ model, we have a random graph with n nodes, with $p = \frac{c}{n}$ denoting the probability that there exists an edge between any 2 nodes.

We set c as a constant to normalize the average degree of nodes as n grows large, allowing us to analyze the structure of a large random graph. Specifically, in a graph with n nodes, each node can connect to at most $n - 1$ other nodes, implying at most $n - 1$ possible edges.

Therefore, the expected degree of a node is:

$$\mathbb{E}[\text{degree}] = (n - 1) \times \frac{c}{n}$$

In the limit $n \rightarrow \infty$:

$$\mathbb{E}[\text{degree}] = \lim_{n \rightarrow \infty} [(n - 1) \times \frac{c}{n}] = c$$

As we have shown in Lectures 2-5, the structure of the graph depends on the value of c :

- if $c < 1$, the graph will contain small components.
- if $c > 1$, the graph will contain a giant component.
- if $c = 1$, we are at the threshold where large components start to appear.

Idea: **Sprinkling** is a probabilistic construction of a graph by gradually exposing the edges of a random graph in rounds to achieve a desired graph property. In our case, we will use Sprinkling to prove the uniqueness of a giant component.

Let's build $G(n, \frac{c}{n})$ in the following 2 steps:

- Construct $G(n, \frac{c'}{n})$ for $c' < c$, which has a lower edge probability (and, therefore, a lower number of edges) than desired. At this stage, the graph does not have a single giant component, but large clusters begin to form.
- If an edge (i, j) is not in $G(n, \frac{c'}{n})$, we sprinkle (i.e. add) it with probability $\frac{\delta}{n}$, where:

$$\delta = c - c' + O(\frac{1}{n})$$

By sprinkling these edges, we transition from $G(n, \frac{c'}{n})$ to $G(n, \frac{c}{n})$. Therefore, the probability that edge (i, j) is not in $G(n, \frac{c}{n})$ is:

$$\mathbb{P}(\text{missing edge in } G(n, \frac{c}{n})) = (1 - \frac{c'}{n})(1 - \frac{\delta}{n}) = (1 - \frac{c}{n})$$

showing that the correct choice of δ is $c - c'$.

Proof of Uniqueness of Giant Component in $G(n, \frac{c}{n})$

Let's define the survival probability of a Galton-Watson branching process with a $Poisson(c)$ distribution, where $c > 1$ and:

$$\Theta(c) = \mathbb{P}(T_{Poisson(c)} \text{ survives})$$

Note that $\Theta(c)$ is the probability that the branching process continues indefinitely, and:

- if $c > 1$, then $\Theta(c) > 0$ and there is positive probability of survival.
- if $c < 1$, then $\Theta(c) = 0$ and the branching process will eventually die out.

Let's also define the set of nodes that belong to large components, $V_+(c)$, as:

$$V_+(c) = \left\{ x \mid |C(x)| > \frac{2c}{(c-1)^2} \log n \text{ in } G(n, \frac{c}{n}) \right\}$$

Step 1)

In order to prove the existence of a giant component, let's start by showing that a large proportion of nodes belong to components of size $\geq \frac{2c}{(c-1)^2} \log n$:

- Fix $\epsilon > 0$ and pick $c' < c$.
- In homework 3, we proved that $\exists c' < c$ such that:

$$|V_+(c')| \geq \left(1 - \frac{\epsilon}{4}\right) \Theta(c')n \geq \left(1 - \frac{\epsilon}{2}\right) \Theta(c)n$$

which implies that a large proportion of the nodes in $G(n, \frac{c'}{n})$ are already part of large (but potentially disconnected) components.

Step 2)

Consider all components $C_1, \dots, C_k$ in the set $V_+(c')$. While we know that they are large components in $G(n, \frac{c'}{n})$, we do not know whether they form a giant component (as some components may have no edges between them). Therefore, let's prove that sprinkling edges almost surely combines the large components into a single giant component.

If sprinkling edges does not connect all components in $V_+(c')$, we can partition $V_+(c')$ (as seen in Figure 2) into 2 disjoint sets, V_A and V_{A^c} where:

$$V_A = \bigcup_{i \in A} C_i \quad \text{and} \quad V_{A^c} = \bigcup_{i \in A^c} C_i$$

and there are no edges between V_A and V_{A^c}

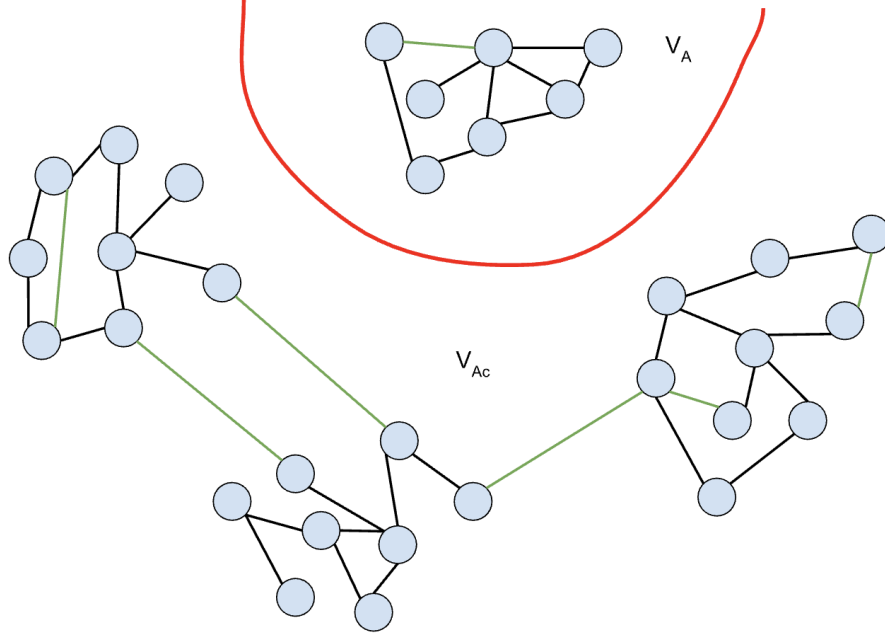


Figure 2: Internal edges in black, Sprinkled edges in green, and red boundary represents partition between V_A and V_{A^c} .

In homework 3, we showed that if:

$$|C_{\max}| \leq (1 - \frac{\varepsilon}{2})|V_+(c')|$$

then both A can be chosen such that V_A and V_{A^c} must have size $\geq \frac{\varepsilon}{2}|V_+(c')|$.

The number of potential edges we can sprinkle is $|V_A| \times |V_{A^c}|$. Since each sprinkled edge is added with probability $\frac{\delta}{n}$, we can compute the expected number of edges between V_A and V_{A^c} as:

$$\mathbb{E}[\text{edges between } V_A \text{ and } V_{A^c}] = |V_A| \times |V_{A^c}| \times \frac{\delta}{n}$$

Therefore, the probability of no edges between V_A and V_{A^c} is:

$$\mathbb{P}(\text{no edges between } V_A \text{ and } V_{A^c}) = (1 - \frac{\delta}{n})^{|V_A| \times |V_{A^c}| \times \frac{\delta}{n}}$$

Since the power becomes exponentially small as $n \rightarrow \infty$, the probability that no edge is sprinkled between V_A and V_{A^c} tends to 0.

Therefore, with high probability, the components will eventually be connected by the sprinkled edges, forming a giant component.

1. Preferential Attachment

In Network Theory, Preferential Attachment is when new nodes attach preferentially to networks (of nodes) that are already well-connected.

Brief History:

- The notion of preferential attachment was first introduced by **Yule** in **1925**, who proposed it to explain why certain species were more likely to grow rapidly in population.
- Then, in **1929**, **Zipf** studied frequency distribution in languages and found that the frequency of k 'th was inversely proportional to k . Zipf's findings shed light on the idea of power-law distributions, where the probability of some large number k occurring is proportional to a power $k^{-\alpha}$.
- In **1955**, **Simon** provided the first rigorous analysis demonstrating power-law distributions through the lens of stochastic processes. He showed that larger categories (i.e. categories with more elements) are more likely to get even larger.
- Finally, in **1999**, **Barabási** and **Albert** developed the BA model to explain power-laws in real networks. The study also popularized the notion of "Powerlaws for the Internet," a phenomenon where a small number of nodes (i.e. websites, networks, etc.) receive disproportionately larger number of connections or traffic. Barabási and Albert demonstrated the existence of preferential attachment in networks like the internet, leading to a "scale-free network."

2.1 Mathematical Definition of the BA Model

- Let's start by looking at an arbitrary finite graph G_1 .
- Then, to transition from a graph at time t (G_t) to the graph at time $t + 1$ (G_{t+1}), we will add a new vertex with m new edges, randomly connecting it to the vertices in G_t .
- The probability that the new vertex v connects to a vertex $u \in G_t$ is:

$$P_u(t) = \frac{d_u(t)}{\sum_{w \in G_t} d_w(t)}$$

- By the Handshaking lemma, the total degree of G_t is:

$$\sum_{w \in G_t} d_w(t) = 2|E(G_t)| ,$$

simplifying $P_u(t)$ to:

$$P_u(t) = \frac{d_u(t)}{2|E(G_t)|}$$

indicating that the probability of the new vertex connecting to an existing vertex u in G_t is directly proportional to the degree of u .

The version of the BA model we covered above was an **Independent Model**, where the m new vertices are connected to G_t **independently** of one another. However, the BA model can be constructed in two additional ways:

- In the **Conditional Model**, we will condition the edges to be connected to different vertices, to ensure that we have at most one edge between the new vertex and an existing vertex.
- In the **Sequential Model**, the new vertex v will choose its first edge to an existing vertex. Then, the degrees of v and the chosen vertex both increase by one, before v chooses the next vertex. We see that degrees of vertices are updated in each step, implying that earlier edge selections affect later selections.

Consider the simple case where G_1 is the graph of 2 vertices connected by m edges, as shown in Figure 3 for the case $m = 3$:

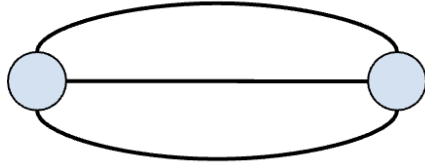


Figure 3: Two vertices connected by three edges

We observe that the number of vertices in the graph at time t , denoted by $|V_t| = t + 1$, which holds as we add more vertices. In a similar way, the number of edges at time t is tm . Therefore, the total degree of G_t at time t is $2mt$, and the probability of selecting a vertex u already in G_t (while constructing edges for a new vertex) is given by:

$$P_u(t) = \frac{d_u(t)}{2mt}$$

2.2 Expected Degrees of Vertices

Now, let's take a closer look at the expected degrees of vertices as t grows large:

- At time $t = 1$: $d_1(1) = d_2(1) = m$
- We can represent the degree of a vertex (in G_t) as:

$$d_{i+1}(t) = d_i(t) + \text{Binomial}\left(m, \frac{d_i(t)}{2mt}\right)$$

since the new vertex can have m possible edges to vertex i where each edge occurs with independent probability $P_i(t) = \frac{d_i(t)}{2mt}$.

- Therefore:

$$E[d_i(t+1) | G_t] = d_i(t) + m \frac{d_i(t)}{2mt} = d_i(t) \left[1 + \frac{1}{2t}\right]$$

stating that the expected degree of vertex i at time $t+1$ (given the graph at time t) is the sum of i 's degree at time t and the expected number of edges it gained with the new vertex.

- Vertices are born at time t_i with degree m except for vertex 1 and 2: $t_1 = t_2 = 1$, $t_3 = 3, \dots, t_i = i \quad \forall i \geq 3$, so:

$$\begin{aligned} E[d_i(t)] &= d_i(t_i) \prod_{s=t_i}^{t-1} \left[1 + \frac{1}{2s}\right] \\ &\approx m \exp\left(\frac{1}{2} \sum_{t_i}^{t-1} \frac{1}{s}\right) \approx m \exp\left(\frac{1}{2} \int_{t_i}^t \frac{ds}{dt}\right) \\ &= m \exp\left(\frac{1}{2} [\log t - \log t_i]\right) \\ &= m \sqrt{\frac{t}{t_i}} \end{aligned}$$

2.3 Degree of a Random Vertex

Assume that vertex i got connected by an edge to the network at time t_i , which tends to the i for large t . Let's assume that vertices arise uniformly between $[0, 1]$, so:

$$t_i \approx i \approx Ut, \quad U \in [0, 1]$$

We are interested in the probability that the degree d_i is greater than the threshold k :

$$P(d_i(t) \geq k) = P\left(m \sqrt{\frac{t}{t_i}} \geq k\right).$$

substituting $t_i = Ut$ gives the following:

$$P\left(U \leq \frac{m^2}{k^2}\right)$$

$$= P\left(U \leq \frac{m^2}{k^2}\right) = \frac{m^2}{k^2}$$

Define $N_k(t)$ as the number of vertices with degree k at time t . It then follows from our probability calculations in above:

$$N_k(t) \sim \frac{2m^2}{k^2}t$$

2.4 Theorems Describing Asymptotic Behavior

1.)

$$\frac{d_i(t)}{\sqrt{t}} \xrightarrow{\text{a.s.}} X_i, \quad \text{r.v. in } \mathbb{R}^+$$

which suggests that, as $t \rightarrow \infty$, the degree $d_i(t)$ of vertex i (when divided by $\sqrt{t}$) converges almost surely to a random variable X_i .

2.)

$$\frac{N_k(t)}{t} \xrightarrow{\mathbb{P}} \rho_k = \frac{2m(m+1)}{k(k+1)(k+2)} \sim \frac{\text{const}}{k^3}$$

where $N_k(t)$ represents the number of nodes with degree k at time t . As $t \rightarrow \infty$, the fraction of degree k vertices converge in probability to ρ_k . From our last simplification, it is evident that the degree distribution of this network follows a power-law distribution (i.e. a scale-free network). So, the larger k values, while being less likely, have relatively high likelihoods, roughly $\frac{1}{k^3}$.

2.5 Proofs for Theorems in section 2.4

Proof for 1.)

To prove convergence of the degree $d_i(t)$ of vertex i (when divided by $\sqrt{t}$), let's show that the normalized degree is a Martingale:

$$M_i(t) = \frac{d_i(t)}{\mathbb{E}[d_i(t)]}$$

If the normal degree is a Martingale, we can use martingale convergence theorems to prove 1.)

In 2.2), we showed that the expected degree $\mathbb{E}[d_i(t)]$ is:

$$\mathbb{E}[d_i(t)] = \prod_{\sigma=t_i}^{t-1} \left(1 + \frac{1}{2\sigma}\right)$$

We can approximate $\mathbb{E}[d_i(t)]$ for large σ as:

$$\mathbb{E}[d_i(t)] \approx \exp\left(\sum_{\sigma=t_i}^{t-1} \frac{1}{2\sigma}\right) \approx \exp\left(\frac{1}{2} \ln\left(\frac{t}{t_i}\right)\right) = \sqrt{\frac{t}{t_i}}.$$

To show that $M_i(t)$ is a martingale, it must satisfy:

$$\mathbb{E}[M_i(t+1) \mid d_i(t)] = M_i(t).$$

Let's start by writing degree $d_i(t+1)$ as:

$$d_i(t+1) = d_i(t) + \text{Bern}\left(\frac{d_i(t)}{2t}\right)$$

since vertex i gains an edge with probability $\frac{d_i(t)}{2t}$.

Then, taking expectation on both sides yields:

$$\mathbb{E}[d_i(t+1)] = \mathbb{E}[d_i(t)] \left(1 + \frac{1}{2t}\right).$$

Therefore, the conditional expectation $\mathbb{E}[d_i(t+1) \mid d_i(t)]$ is:

$$\mathbb{E}[d_i(t+1) \mid d_i(t)] = d_i(t) \left(1 + \frac{1}{2t}\right)$$

Recall that $M_i(t+1)$ is defined as:

$$M_i(t+1) = \frac{d_i(t+1)}{\mathbb{E}[d_i(t+1)]}$$

Now, let's take the conditional expectation of $M_i(t+1)$, giving us:

$$\mathbb{E}[M_i(t+1) \mid d_i(t)] = \frac{\mathbb{E}[d_i(t+1) \mid d_i(t)]}{\mathbb{E}[d_i(t+1)]}.$$

After substituting the conditional expectation $\mathbb{E}[d_i(t+1) \mid d_i(t)]$, we are left with:

$$\mathbb{E}[M_i(t+1) \mid d_i(t)] = \frac{d_i(t) \left(1 + \frac{1}{2t}\right)}{\mathbb{E}[d_i(t+1)]} = \frac{d_i(t) \left(1 + \frac{1}{2t}\right)}{\mathbb{E}[d_i(t) \left(1 + \frac{1}{2t}\right)]}$$

Finally, we are left with:

$$\mathbb{E}[M_i(t+1) \mid d_i(t)] = \frac{d_i(t)}{\mathbb{E}[d_i(t)]} = M_i(t)$$

$M_i(t)$ is an L^1 martingale (i.e. $\mathbb{E}[|M_i(t)|] < \infty$). The convergence of the L^1 martingale $M_i(t)$ then implies that:

$$M_i(t) = \frac{\mathbb{E}[d_i(t)]}{\mathbb{E}[d_i(t)]} \xrightarrow{\text{a.s.}} \tilde{X}_i$$

statement 1 now follows since $\frac{\mathbb{E}[d_i(t)]}{\sqrt{t}}$ converge, giving $X_i = \tilde{X}_i \lim_{t \rightarrow \infty} \frac{\mathbb{E}[d_i(t)]}{\sqrt{t}}$

Proof Outline for 2.)

First, we will show that the expected degree distribution of the network converges as t grows large, showing that

$$\frac{1}{t} \mathbb{E}[N_k(t)] \rightarrow \frac{2m(m+1)}{k(k+1)(k+2)}$$

Next, we want to show that $N_k(t)$ is concentrated around its expectation, which we can show with high probability using a Martingale. Let's first introduce G_t which represents the network at time t . The choice of vertices that receive new edges will be denoted by $Y_t = (v_{t1}, \dots, v_{tm}) \in V(G_t)$.

Let's define the conditional expectation of $N_k(t)$ given the choice of vertices to receive new edges up to time s as:

$$M_s = M_s(Y_1, \dots, Y_s) = \mathbb{E}[N_k(t) \mid Y_1, \dots, Y_s]$$

Let's also express the number of vertices with degree k at time t and initial expectation in Martingale notation:

$$M_0 = \mathbb{E}[N_k(t)] \quad M_t = N_k(t)$$

Now, to show that M_s is a Martingale, we must show that it has the Martingale property:

$$\begin{aligned} \mathbb{E}[M_s \mid Y_1, \dots, Y_{s-1}] &= \mathbb{E}[\mathbb{E}[N_k(t) \mid Y_1, \dots, Y_s] \mid Y_1, \dots, Y_{s-1}] \\ &= \mathbb{E}[N_k(t) \mid Y_1, \dots, Y_{s-1}] = M_{s-1} \end{aligned}$$

Repeat the Martingale construction for the Martingale M_s . Then we can bound the Martingale differences as:

$$|M_s - M_{s-1}| \leq 2m$$

To prove this concentration, we must use the Azuma-Hoeffding inequality.

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